Project 3 — Design Direction & Research Compilation

SECTION 1: Person

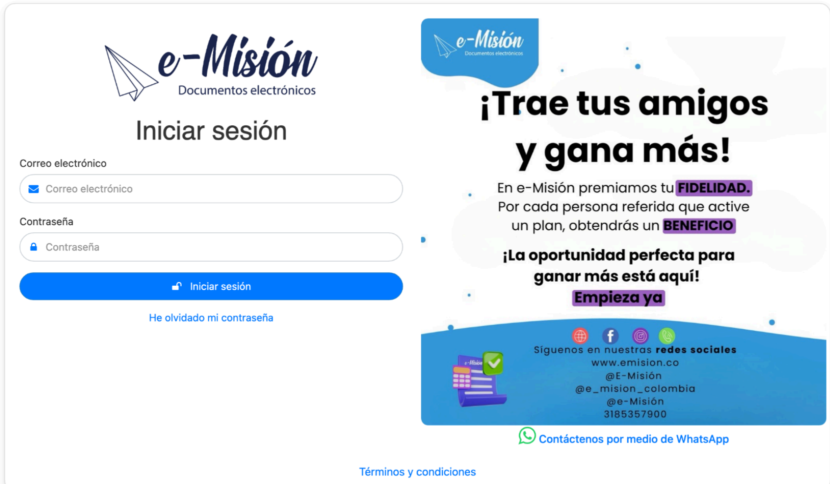
- **Name:** Rodrigo Múnera Vélez
- **Relationship to you:** My father, someone I talk to every single day
- **Profession:** Plastic surgeon practicing in Colombia
- **Practice profile:** Independent practitioner; sees approximately 15 new patients per month; patient base skews heavily female; works across multiple clinics but mainly at IQ InterQuirófanos located in El Poblado, Medellín, Colombia.
- **Access for the project:** Direct and frequent, I have verbal consent for project participation, use of his branding, and presentation of his information in class.
- **Branding/identity:** Has a defined visual identity: circular monogram logo, deep navy (#172137) and white palette, clean geometric uppercase typography for his name with the descriptor "Cirujano Plástico"
- **Instagram presence:** @dr.rodrigomunera

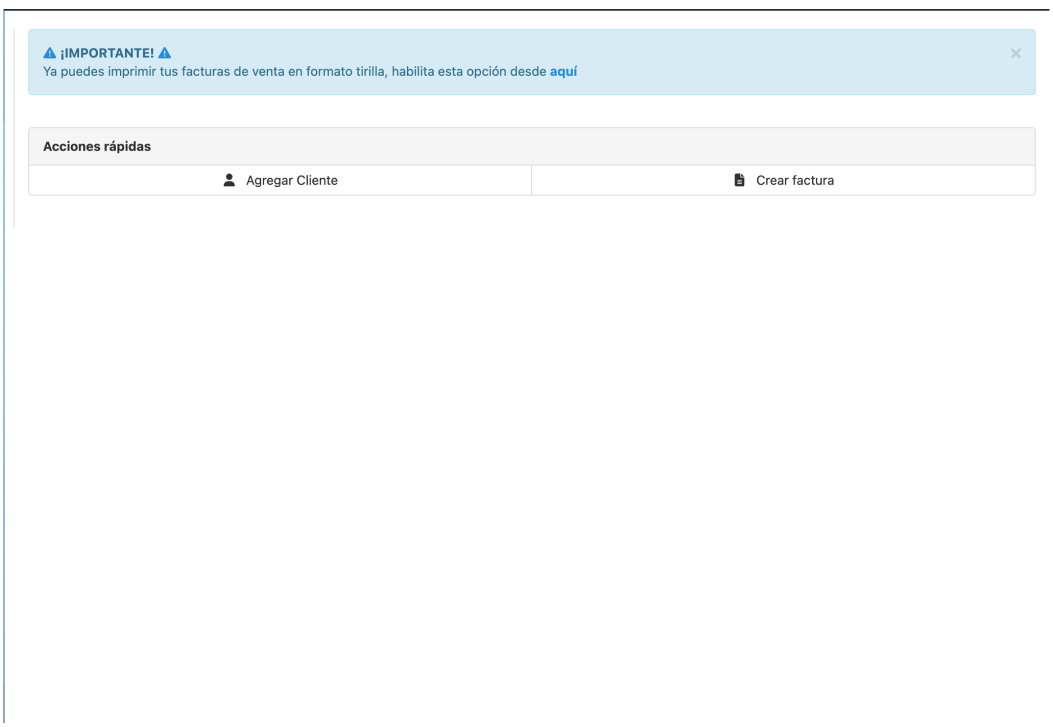
I chose to build a product for my father because I experience first hand the frustration when it comes to technological work. Moreover, developing this product will buy him more time to spend on the things that truly matter, like spending time with my sisters, his girlfriend, and myself. I am the right person to design for him because I know exactly how the process and his workflow functions. I am in charge of a big portion of his administrative work and I understand exactly how it must be done by government regulations, and his own criteria. I know exactly what he needs and how to create real speed for his workflow.

SECTION 2: The Problem

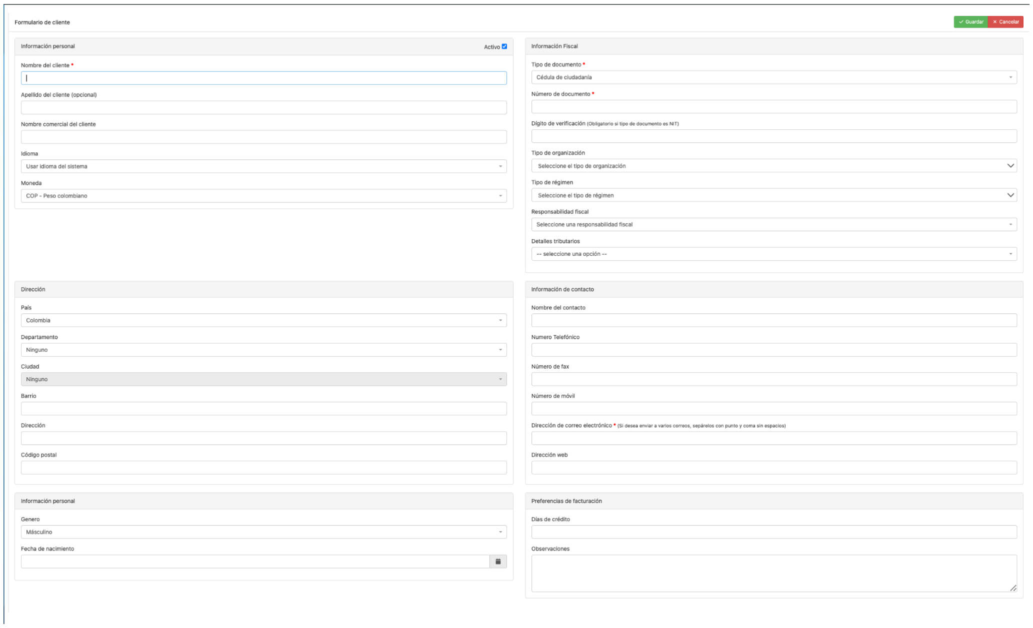
Relatively recent Colombian regulation requires independent medical professionals to manage their own electronic billing (facturación electrónica). This is something that was formerly handled by clinics or accountants. This means he must now register every new patient as a Cliente and create their invoice himself, on the e-Misión platform (`facturador.emision.co`).

The platform's workflow (Screenshots and explanations provided by my father, translated with AI):

1. Log in to facturador.emision.co
    a. 

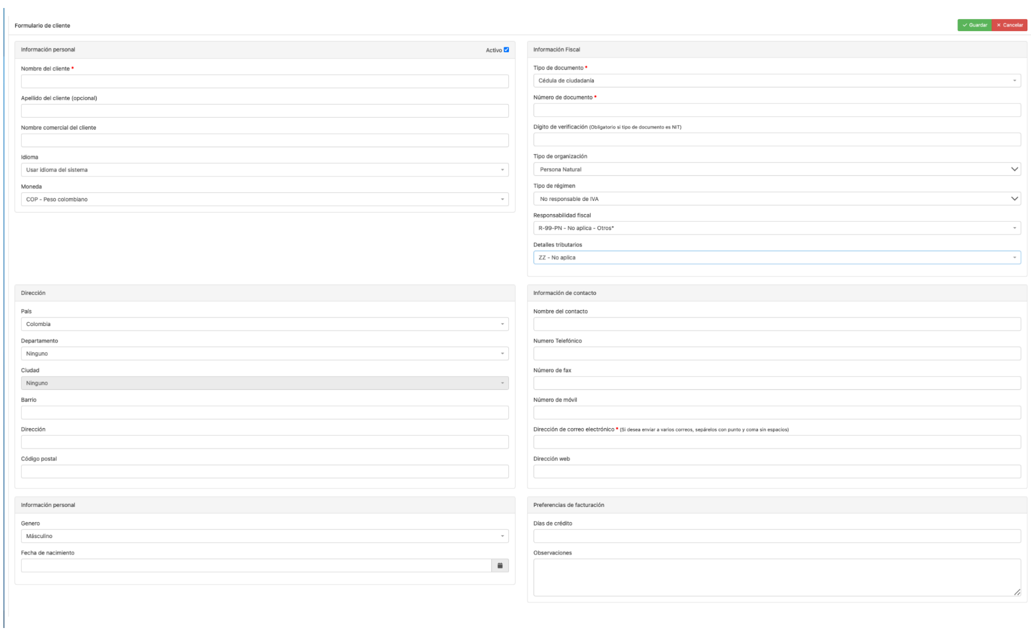
2. Click "Agregar Cliente"
    a. 

3. Manually fill ~10 text fields (Nombre, Tipo de documento, Número de documento, Departamento, Ciudad, Dirección, Género, Teléfono, Email)



a.

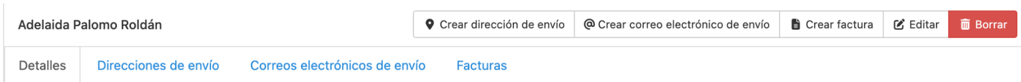
4. Select 4 default dropdowns that are the same every single time: Persona Natural / No responsable de IVA / R-99-PN – No aplica – Otros* / ZZ – No aplica



a.

5. Save the client

a.

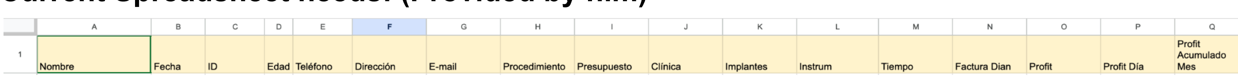
6. Click "Crear factura" → confirm dropdowns → send
    a. 

7. (Step 6 currently blocked because he doesn't yet have a resolución vigente from DIAN, so v1 of your tool targets steps 1–5 only)
    a. 

Data source: The information needed for those fields exists on physical printed clinic history forms that each patient fills out at intake. The patient's email is handwritten at the top of the printed form; the other fields are printed.

Current workflow observed: My dad maintains a yearly Excel spreadsheet (`Pacientes_2026.xlsx`) where he himself re-types the patient data from the paper forms. Then, when it's time to bill, the data must be re-typed *again* from Excel into e-Misión. I am always in charge of doing this second copy-paste step, which is how I came to know this problem so intimately.

Current Spreadsheet needs: (Provided by him)



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Nombre	Fecha	ID	Edad	Teléfono	Dirección	E-mail	Procedimiento	Presupuesto	Clínica	Implantes	Instrum	Tiempo	Factura Dian	Profit	Profit Día	Profit Acumulado Mes

Main pain points:

1. He has to enter into the website... and enter each patient manually, which is a process I usually do by copying and pasting.
2. I need to build something extraordinary that significantly reduces his admin work time.
3. The copy-paste flow is essentially what he already does switching tabs between his spreadsheet and the platform. This wouldn't really be a major change in his current workflow. Hence he needs an automatization that helps him significantly reduce the time he spends doing these administrative tasks.

Volume: ~15 patients per month × ~5 minutes of typing/clicking per patient on emisión = 45-60 minutes of pure administrative friction every month, separate from the time spent populating the spreadsheet in the first place.

The current workarounds in his life that reveal what's broken:

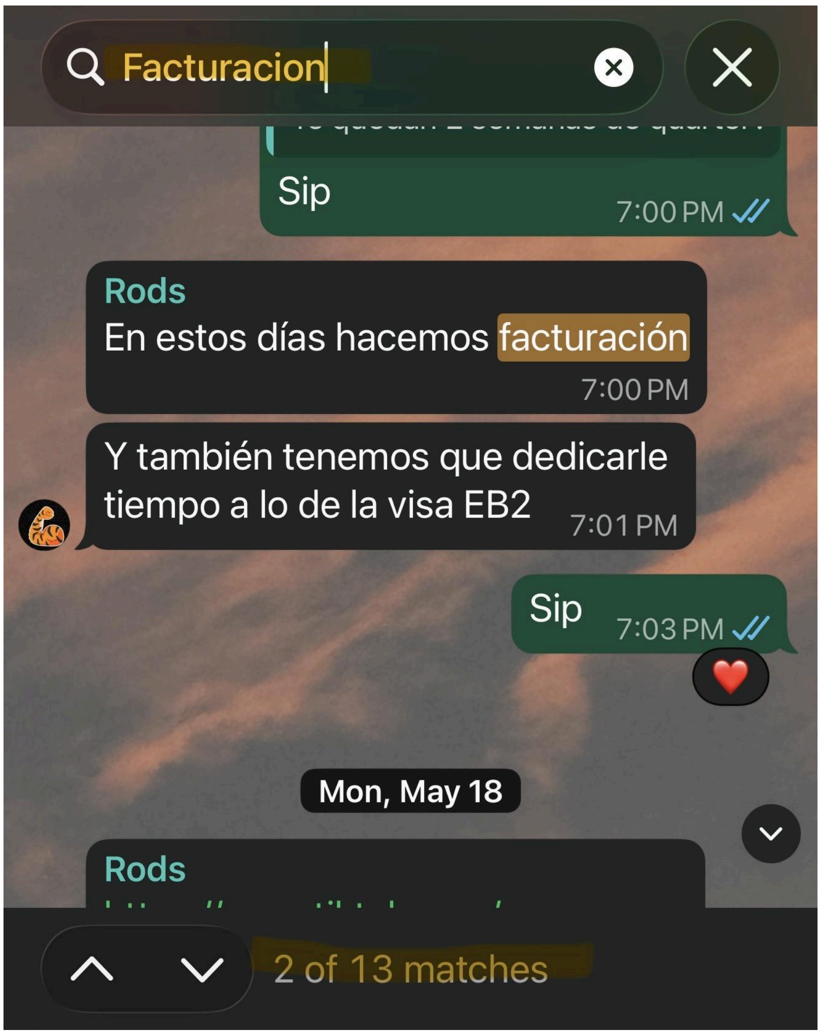
- Me acting as a human RPA system between his Excel and the government portal
- A paper-first patient intake process that already captures all the necessary data
- A spreadsheet workflow that captures everything correctly but doesn't talk to anything downstream
- Manual department lookups by Googling Colombian cities when only a partial address is on the form

Problem Statement:
The current electronic billing process forces independent medical professionals to manually re-enter patient information across disconnected systems, creating time-consuming administrative work. While my father still collects and stores accurate patient data through paper intake forms and spreadsheets, the lack of automation between these workflows and the e-Misión platform creates unnecessary wasted time that takes time away from patient care.

SECTION 3: Research Documentation

Evidence	**What it shows**	**Where it lives**
`Pacientes_2026.xlsx` (real file)	His ledger structure, 17 columns, ~1010 pre-formatted rows for the year, no formulas, single sheet `Hoja 1`	I have access to it from my personal gmail and his; goes in repo (de-identified: I have removed the existing patient rows before committing)
6 emisión screenshots	The exact form I fill out, field-by-field, with defaults (drop-down menus).	In repo
Annotated walkthrough of the emisión process	Step-by-step instructions written from doing the process myself	README
My dad's logo (logo.ai, logo2.ai, logo2.pdf)	Branding direction for the app	PNG render of logo2.pdf used to extract navy hex

Evidence	What it shows	Where it lives
My dad's Instagram (@dr.rodrigomunera)	Public identity + branding	Public URL, can be referenced
Colombian regulation context	Frames why this problem exists in the first place	DIAN's electronic invoicing mandate in this document
Chats with my dad	Pain points, need for the app, etc.	In this document.



Every month for over a year, he's asked me for help with his electronic billing, a process that not only requires a significant amount of his time, but of which he fully depends on me to do for him. The thing that he liked the most when I first proposed the idea is the fact that he'll no longer need my help when it comes to this task, as the process will only take him 5 minutes in a very user-friendly way.

SECTION 4: Definition of "Helped"

Success Metrics and flow:

- Photograph a paper form, see the data extracted automatically, confirm it, and have it appear in his spreadsheet without typing.
    - Still being able to type in the spreadsheet when needed to correct or add data.
- When he sits down to do emisión, the Chrome extension fills every field correctly — including dropdowns, so he just visually confirms and hits "Guardar."
    - He no longer needs you as the copy-paste machine between his spreadsheet and emisión
- The 45-60 minutes/month of administrative friction must drop to under 10 minutes
- His existing Excel structure must stay exactly the same as he knows it (with three small, agreed-upon additions)

Success means completely erasing the administrative delays from his monthly routine, reducing a 45-to-60-minute task down to under 10 minutes. By automating data extraction directly from a photo into his existing Excel structure, he eliminates manual typing and no longer needs an intermediary to copy-paste data into emisión. Finally, success means that he can gain full independence in this process and will no longer need me to help him monthly, requiring less time and clicks.

SECTION 5: My Qualification

Why am I the right person to complete this task?

- I am his daughter meaning that I keep daily contact, have direct access, and can perform frequent observations.
- I currently do the work I'm proposing to automate, meaning I have first-person knowledge of the friction.
- I can read his handwriting, recognize his clinic's form, know his terminology, speak his language (Spanish), understand the cultural context of Colombian medical billing.
- I have a designer's eye for his existing brand and I know what would feel "his" vs. what would feel generic.
- I'm in a course teaching me to direct AI for production work plus my User Experience Design major program, meaning I have the skills to develop this product.

I am the only person who can build this because I have first-hand operational knowledge, specialized technical skill, and deep personal trust. As his daughter, I have daily access and an intimate understanding of his workflow, but more importantly, I am the one currently doing all of the manual process. Additionally, combined with my UX design background and AI production, I have the unique capability to translate this frustration into a high-fidelity solution. I speak his language, understand his handwriting, know the Colombian medical billing process, and hold a designer's eye for his brand because I created it. Anyone else would just be building software, whereas I am building an extension of his daily practice that no outsider could replicate.

SECTION 6: Platform Decision and Reasoning

1. **A Progressive Web App (PWA)** — React, deployed on Vercel, mobile-installable. Runs on his iPhone 17 Pro Max for clinical-hours capture, on his iPad for review, on his Mac for desk work. Handles: photographing the paper form, extracting the data via Claude API, reviewing/correcting, appending to the Excel.

2. **A Chrome extension** (sideloaded in developer mode, no Chrome Web Store) — runs only on `facturador.emision.co`. Reads the queue of pending patients from shared local storage. Fills every field on the Agregar Cliente form, including the four standard dropdowns. Stops before Guardar so he visually confirms.

Why two surfaces:

- Capture happens at clinical hours, on a phone, with paper on the other hand. A desktop browser can't reasonably photograph paper.
- emisión submission happens at a desk, in a Chrome browser. An iPhone Safari extension can't manipulate a web form the way a desktop Chrome extension can.
- These are two different contexts in his life. One surface would force a compromise that fails both contexts.

Why React (PWA), not native iOS:

- Single codebase that works on his iPhone, his iPad, his Mac, his iCloud
- No app store review process meaning I control deployment cadence completely
- "Add to Home Screen" gives him a real app icon without distribution friction
- React + Vite + Tailwind is the assignment's recommended stack and my skills carry forward

Why a Chrome extension (sideloaded), not Chrome Web Store:

- He is the only user. No third party needs access.
- Sideloading skips Chrome Web Store review (1-3 days)
- Internal-tool sideloading is standard industry practice for personalized automation

Why not a backend database / cloud sync:

- Patient data should not be stored on any server we control
- 15 patients/month volume doesn't justify backend complexity
- iCloud Drive export/import handles cross-device needs at this volume

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SECTION 7: Non-Negotiables

1. **The app must be in Spanish.** Every label, every error, every confirmation. He is the sole user; his language is the only language.
2. **His Excel template must be preserved exactly.** The only additions are the three he agrees to (Tipo de doc, Género, Resumen sheet). His existing column order, naming, formatting must be untouched.
3. **No real patient data can pass through any cloud API for the class deliverable.** Fake/de-identified test data only.
4. **The tool must never submit unverified data.** Every extracted field is reviewable; low-confidence fields are visually flagged; he confirms before any row is added or any form is submitted to emisión.
5. **The visual design must feel like a tool designed specifically for him.** All of his branding including his navy blue, his monogram, his name, his vocabulary, his implicit aesthetic should be preserved.
6. **It must work on his devices**: iPhone 17 Pro Max, iPad, Mac in Chrome.
7. **Time saved per session must be substantial enough to feel like transformation.** Should reduce at least 6 times the manual entry work.
8. **The class deliverable must be defensible against the Five Questions.** Every design decision traces to either his observed need or my explicit research finding.

SECTION 8 — Five Questions Reflection

The brief lists five questions. They hit different in Project 3 because there's a real person on the other end. I cannot answer them for you. But here's what each one probes, so you know what to write about:

1. **Can I defend this?**
    - Yes, I can defend the decisions I made because almost every important feature came directly from something I observed in my father's real workflow. The clearest example is the Chrome extension. At first, AI suggested a clipboard-based system where patient data would simply be copied from one

place into another, but I immediately knew that would not actually solve the problem because I had personally been the person doing that copy-paste work for him for months. It would have been the exact same process with slightly different screens. Another strong example is the “Resumen” sheet. That was not added as a convenience feature, but my father explained that the monthly “Factura Dian” total is what he uses to calculate his Colombian seguridad social payments, including pension, salud, and ARL contributions. This is something that made the sheet financially and legally important. Smaller decisions also came from direct observation: I removed automatic date defaults because he often bills patients days or weeks after surgery, I kept the interface entirely in Spanish using Colombian medical billing terminology because he is the only user, and I chose local storage with no backend because patient information is protected under Colombian habeas data laws. Even the speed optimization came from testing with him directly: the original 30-second extraction time was simply too slow for clinical use, so I redesigned the pipeline until it reached around 4–8 seconds. Looking back, I think what I would improve is documenting these observations more formally earlier in the process instead of relying mostly on ongoing conversations and hands-on involvement.

2. **Is this mine?**
    - I believe this project is genuinely mine because the important decisions did not come from AI and they came from my understanding of my father’s life and workflow. The strongest example was when AI suggested a clipboard-based solution for transferring patient data into emisión. I rejected it almost immediately because I knew, from firsthand experience, that this was already the exact process we were trying to move away from. I had personally spent months switching between spreadsheets and the platform copying patient information field by field, so I knew that solution would not meaningfully reduce his workload. The same thing happened with the automatic date selection. AI assumed the “today” date would be correct because that is standard UX logic, but I overrode it because I knew my father usually photographs patient forms during surgery days and handles billing later. Another important moment was when AI assumed the Excel system used separate monthly sheets, but once I introduced the real spreadsheet, that assumption turned out to be wrong. I think this process taught me that directing AI is about constantly evaluating whether its suggestions actually match the reality of the person you are designing for. Finally, I had a clear role in filtering every decision through what I knew about my father’s real needs.
3. **Did I verify?**
    - Yes, the project was verified continuously throughout development because my father was actively involved in shaping the system as it was being built. I did not disappear for weeks and then showed him a finished prototype at the end. Instead, he regularly supervised decisions through conversations, testing sessions, and feedback while I was developing the tool through phone calls. He

watched his actual Pacientes_2026.xlsx file load correctly into the app with patient information, confirmed that the extraction speed became acceptable once it reached around 4–8 seconds, and directly requested features like multi-select after realizing he often processes patients in batches for emisión submission. He also watched the Chrome extension fill the real "Agregar Cliente" form on the emisión platform using his patients' data, which was one of the clearest moments of validation in the project. Another important piece of feedback came when he identified the need to capture clinical information like Procedimiento and Factura Dian during the same workflow as the patient photo. I think one thing I could improve in the future would be creating a more formal testing structure with documented task observations instead of relying mostly on ongoing collaborative feedback. Still, I honestly think his continuous involvement made the verification process stronger because the tool evolved directly around his daily routine instead of being evaluated only after completion.

4. **Would I teach this?**
    - Yes, I think I understand this project well enough to explain and teach the main architectural decisions behind it. I can clearly explain why the system became a two-surface architecture instead of a single app: my father's workflow naturally happens across two contexts. Patient intake and photo capture happen quickly at the clinic on mobile, while billing happens later at a desktop computer inside emisión. Trying to force both tasks into one interface would have created new forms of friction. I can also explain the extraction pipeline in detail, how the images are compressed client-side before being sent to a Vercel serverless function connected to the Anthropic API, and how the extracted patient information is returned as structured JSON with confidence scoring. I understand the privacy reasoning behind keeping the API key in environment variables and using local storage instead of maintaining a backend database with sensitive patient information. I also understand the logic behind the Excel export system and the Chrome extension's DOM-based autofill process. At the same time, there are still some technical areas where I would probably need notes before teaching them confidently, especially lower-level implementation details like certain SheetJS syntax or IndexedDB structures. I think I understand the system very strongly from a product design and systems-thinking perspective, even if there are engineering details I am still learning more deeply.
5. **Is my disclosure honest?**
    - Yes, I believe my disclosure is honest because the AI Direction Log and Records of Resistance describe real events that happened during development. Some of the strongest moments in the documentation came directly from real conversations and real frustrations I experienced while helping my father with billing. The documentation itself was generated with AI, which my professor explicitly allowed, but the decisions, resistance moments, and workflow observations it describes are genuine. My father supervised the entire project throughout development, and many of the final features exist because of his

direct feedback. I also think it is important to be honest about the fact that the process was sometimes messier and less linear than the logs make it appear. Many decisions changed through testing and conversation. Still, I stand behind the overall picture the logs present because they accurately reflect the relationship between myself, the AI tools, and my father's real workflow. More than anything, this project was about using AI to help build something specifically around one person's everyday reality.

Section 9 — Observations during testing (Paraphrased, all testing was made in Spanish)

- My dad was able to separate the two moments without needing instructions. This confirmed the decision to split the system was on the right track.
- During testing, he hesitated when the extraction process originally took around 30 seconds and commented that it felt too slow for something he would use repeatedly between patients. Once the processing time dropped to around 4–8 seconds, he became more comfortable continuing through the workflow.
- When reviewing extracted patient data, he consistently checked ID numbers, patient names, and email first before looking at any other field. These are the most important categories in which the system must work with high accuracy.
- When I observed him process multiple patients, it was clear that he grouped billing into batches based on the month. This directly led me to the addition of the multi-select feature in the Registry view for faster emisión submission.
- He showed frustration whenever a workflow required unnecessary repeated clicking or tab switching, especially when moving between the main spreadsheet and the emisión platform. This confirmed to me the idea that reducing interaction repetition was way more important to him than adding additional features or visual complexity.
    - He is not that tech savvy, meaning that this tool must be developed in the simplest way possible.
- When testing the Chrome extension, he reacted most positively to seeing the emisión fields fill out automatically in real time.
- He consistently processed billing from a desktop computer even after successfully using the mobile capture flow.
    - This observation confirmed that the desktop version is still essential because emisión itself is significantly easier to manage on a larger screen.
- During feedback conversations, he focused far more on reliability and speed than aesthetics. His comments were usually about whether the system reduced administrative effort time, whether the data was accurate, or whether the process felt trustworthy enough for real patient data.

Final User Satisfaction After Final Test

Translated transcript to English:

> “Hello, Vale. Uh, thank you so much for the creation you made with artificial intelligence for electronic billing. It truly makes my life much easier, it helps me save a lot of time, since I no longer have to be writing, typing, looking at a history and typing in Excel, uh, the address, the phone number, the identity document, uh, it saves me a lot of time, it makes things much easier for me. I don't have to strain my eyes, uh, I can do it anywhere, from my cell phone, I don't have to wait until I get home to look for a computer to do it. So, anyway, I'm very grateful, my love, for that creation you made, and that tool makes my life much easier and saves me a lot of time. Thank you very much”

SECTION 10 — Post-Mortem

The development of MediBill RMV, showed me how important it is to design around real-life workflows instead of relying on generic AI assumptions. Throughout the process, the AI kept suggesting common SaaS and mobile app patterns, however, many of these ideas did not actually fit the way the clinic operates day to day and the workflow of what my dad has to complete.

Because I understood the workflow firsthand, the project finally transformed into a dual-system solution: a mobile PWA for taking photos of patient forms at the clinic and a Chrome extension that automatically fills patient data into the emisión platform on desktop, matching Dr. M’s actual routine of working across multiple devices. The extraction pipeline was also heavily optimized, reducing processing time from around 30 seconds to just 4–8 seconds through image compression and a lighter AI model. Even the Excel structure was redesigned around Colombian administrative realities, especially the “Resumen” sheet used to calculate monthly seguridad social payments. In the end, the project showed me that building effective AI tools requires a deep understanding of the user’s environment and real operational constraints.
